

Guarding a city from the COVID-19 pandemic



As of May 5, 2020, the coronavirus disease 2019 (COVID-19) outbreak had spread to 215 countries, areas, or territories and led to more than 3·4 million confirmed cases worldwide,¹ within just 4 months of the first reported cases. The global pursuit of antiviral treatments and vaccines is in progress, but unfortunately none have proven to be effective yet. Without effective treatments, we have to rely on public health interventions to fight the pandemic, by controlling the source of infection, blocking the transmission route, and preventing new infections.²

The foremost issues are how to isolate patients suspected to have COVID-19 from the population and further recognise the confirmed cases. In *The Lancet Digital Health*, Bei Mao and colleagues describe a screening strategy for COVID-19 in fever clinics in Shanghai, China.³ Patients with a fever or with respiratory symptoms and suspected exposure history to an individual with COVID-19 were referred to the fever clinics during hospital visits. All individuals who attended the fever clinics had their medical history taken and were given preliminary laboratory tests and chest imaging if necessary. The identified suspected cases were then given at least two RT-PCR tests for final confirmation. Among the 53 617 patients who sought medical care at the 25 fever clinics enrolled in this study, 1004 patients (1·9%) were considered as suspected cases, and 188 patients (0·4% of all patients, 18·7% of suspected cases) were eventually diagnosed as confirmed cases. The confirmed patients were subsequently transported to designated hospitals for isolation and treatment.

Fever clinics were initially established in China for isolation and triage during the SARS outbreak in 2003.^{4,5} They have played an important role in surveillance and treatment of patients with acute respiratory infections both in epidemic settings (eg, 2009 H1N1 influenza) and during seasonal influenza outbreaks. The fever clinics aimed to reduce within-hospital transmission of COVID-19. They have several advantages compared with traditional emergency departments or out-patient clinics. Fever clinics are separate hospital areas working semi-independently to receive and isolate febrile patients, with examination rooms, essential laboratory and imaging facilities, and overnight observation rooms

all included in one comprehensive unit.⁵ The clinics are equipped with trained medical staff wearing full personal protective equipment. In ideal situations, fever clinics are built with *san qu liang tong dao*, meaning three zones and two passages in Chinese, which is a design often used in isolation wards for infectious diseases to prevent cross-contamination.^{6,7} Suspected cases can be isolated on the spot without patient transfer. In addition, these symptom-oriented clinics often have complete and standard screening protocols to follow during epidemics.

Distinguishing COVID-19 from other illnesses early in the infection lifecycle is not an easy task, especially when the pandemic partly overlapped with the winter influenza season. In the study, bacterial pneumonia, common cold, and influenza were common causes for suspected COVID-19 cases, which were eventually excluded by RT-PCR tests. Therefore, these conditions should be carefully considered in differential diagnosis. The authors assessed risk factors for COVID-19 on the basis of patient data, and reported that epidemiological exposure history, decreased lymphocyte count ($<0\cdot8\times10^9/L$), decreased white blood cell count ($<4\times10^9/L$), ground glass opacity, and having both lungs affected on chest CT, as well as other extrapulmonary presentations (eg, fatigue) were independent risk factors for COVID-19 diagnosis in the suspected population in fever clinics. On the basis of these findings, we look forward to prediction models or triage algorithms being developed for COVID-19 in future studies.

Regardless of the many adjuvant tools available, RT-PCR tests remain the gold standard for confirmation or exclusion of the COVID-19 diagnosis. The capacity for nucleic acid tests is important for screening and epidemic control. The false negative issue of RT-PCR tests remains controversial. It is reported that in the 188 confirmed patients in the study, only 72% had an initial positive nucleic acid test.³ Two of the 188 patients did not get a positive result until their fourth test, indicating the necessity for close monitoring and follow-up in clinically suspected cases.

Shanghai, a large city with more than 25 million permanent residents and nearly 10 million non-permanent residents, has shown efficient control

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of the epidemic during this outbreak thus far. Only 655 confirmed cases have been reported in Shanghai as of May 3, 2020, and the local transmission of the virus has remained low.⁸ The fever clinic screening strategy is not the only guardian of the city. The defence included joint public health interventions from the government, hospitals, and the community. Although at large social and economic costs, these public health interventions (eg, intercity travel restrictions, early identification and isolation of cases, contact restrictions, and social distancing) were modelled to have reduced confirmed case numbers by more than 100 times.^{9,10} There were also additional measures, including daily reports on the epidemic situation, a QR code for citizens to report their travel history and health information, reduction of crowding and mass gatherings, no contact deliveries, community management, and designated hospitals, among others, which were all believed to provide the city with an extra layer of protection.

Controlling the COVID-19 epidemic in a densely populated city is a difficult task. This is especially true in winter and spring when other respiratory pathogens are co-circulating in community. What we have learned from the experience in Shanghai is that standardised screening (early recognition) and sufficient nucleic acid tests (early diagnosis) were prerequisites for the effective treatment of COVID-19, and were key to prevention of spread and control of the epidemic. The methods used in Shanghai might serve as a valuable model for other cities worldwide.

We declare no competing interests.

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